

Multiple Choice (10 marks)

1. How many homomorphisms are there of $\mathbb{Z}$ into $\mathbb{Z}_2$?
 - (a) 1
 - (b) 2
 - (c) infinitely many
 - (d) 0
2. Let G denoted the set of all $n \times n$ non-singular matrices with rational numbers as entries. Then under multiplication G is a/an
 - (a) subgroup
 - (b) finite abelian group
 - (c) infinite, non abelian group
 - (d) infinite, abelian
3. Find the order of the factor group $(\mathbb{Z}_4 \times \mathbb{Z}_{12})/(\langle 2 \rangle \times \langle 2 \rangle)$
 - (a) 2
 - (b) 3
 - (c) 4
 - (d) 12
4. In the complex z -plane, the set of points satisfying the equation $z^2 = |z|^2 i$
 5. pair of points
 6. circle
 7. half-line
 8. line

Statement 1 | The symmetric group S_3 is cyclic. Statement 2 | Every group is isomorphic to some group of permutations.

- (a) True, True
- (b) False, False
- (c) True, False
- (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The homomorphic image of a cyclic group is always cyclic.
7. True or False: Every function from a finite set onto itself is necessarily one-to-one.
8. True or False: The product of the polynomials $f(x) = 4x - 5$ and $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$ is $2x^2 + 5$.
9. True or False: The maximum possible order for an element of $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$ is 240.
10. True or False: A group G in which $(ab)^2 = a^2 b^2$ for all a, b in G must be of order two.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The point $P = (-1, 0)$ lies on the ellipse $4x^2 + y^2 = 4$. Let Q be another point on this ellipse, and let d be the maximum length of $\overline{PQ}$. Find d^2 .
12. Discuss the implications of a linear transformation P on a finite-dimensional real vector space V such that $P^2 = P$. Analyze whether P is necessarily diagonalizable and justify your reasoning.

End of Paper